Use Case Flow Document

Vehicle Parking Management and Tracking System

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UC-01 — Register / Login

Goal	Authenticate a user by allowing them to register an account and log in with valid credentials.
Actor	Driver
Main flow	1. User opens the system and selects registration. 2. User enters required registration information. 3. System validates the information and creates the account. 4. User enters login credentials. 5. System authenticates the credentials. 6. System displays the user dashboard.
Alternate Flow	Invalid Registration Details At Step 3, if validation fails: 3a. System displays validation errors and does not create the account. 3b. User corrects the details and resubmits. Invalid Login Credentials At Step 5, if authentication fails: 5a. System displays an error message and asks the user to retry. 5b. User may retry login or cancel. Account Locked At Step 5, if 5 consecutive failed attempts occur: 5a. System temporarily locks the account. 5b. System notifies the user and blocks further attempts until unlock (IR-SEC-04).

UC-02 — Manage Vehicle

Goal	Allow a driver to register and maintain their vehicle information.
Actor	Driver
Main flow	1. User selects vehicle management. 2. User enters vehicle details (registration number, type, model, color). 3. System validates the information. 4. System checks whether the vehicle already exists. 5. System stores and displays the registered vehicle.
Alternate Flow	Duplicate Vehicle At Step 4, if the vehicle already exists:

	4a. System rejects the entry. 4b. System notifies the user that the vehicle is already registered. Invalid Vehicle Details At Step 3, if validation fails: 3a. System displays a validation error. 3b. User corrects the details and resubmits.

UC-03 — View Parking Availability

Goal	Allow users and staff to view the current availability and status of parking slots.
Actor	Driver / Attendant / Administrator
Main flow	1. User opens parking availability. 2. System retrieves parking-slot information. 3. System identifies slot status. 4. System displays Available, Occupied, Reserved, and Unavailable slots.
Alternate Flow	Data Retrieval Failure At Step 2, if slot data cannot be retrieved: 2a. System displays an error message. 2b. System allows the user to retry. No Matching Slots At Step 4, if a filter is applied and no slots match: 4a. System displays a 'no slots found' message.

UC-04 — Reserve Parking Slot

Goal	Allow an authenticated driver to reserve an available parking slot for a specified period.
Actor	Driver
Main flow	1. User views available slots. 2. User selects a slot. 3. User specifies the reservation period. 4. System verifies slot availability. 5. System creates the reservation. 6. System changes slot status to Reserved and confirms.
Alternate Flow	Slot No Longer Available At Step 4, if the slot is no longer available: 4a. System informs the user. 4b. System asks the user to select another slot. Invalid Reservation Period At Step 3, if the end time is earlier than the start time:

	3a. System displays a validation error. 3b. User re-enters a valid period. Reservation Limit Exceeded At Step 4, if the user already has the maximum allowed active reservations: 4a. System rejects the new reservation. 4b. System informs the user of the limit.

UC-05 — Record Vehicle Entry

Goal	Record a vehicle's entry into the parking facility and open an active parking session.
Actor	Parking Attendant
Main flow	1. Attendant enters the vehicle number. 2. System retrieves vehicle information. 3. System checks for an existing active session. 4. System checks slot availability. 5. Attendant confirms entry. 6. System creates a parking session, records entry time, assigns a slot, and marks it Occupied.
Alternate Flow	Active Session Already Exists At Step 3, if the vehicle already has an active session: 3a. System rejects the entry. 3b. System displays an error (DR-02). No Suitable Slot Available At Step 4, if no suitable slot is available: 4a. System informs the attendant. 4b. Entry is not processed until a slot becomes available. Unregistered Vehicle At Step 2, if the vehicle is not found: 2a. System prompts the attendant to register the vehicle first.

UC-06 — Assign Parking Slot

Goal	Assign an appropriate available parking slot to an entering vehicle.
Actor	Parking Attendant
Main flow	1. System retrieves available slots. 2. System identifies suitable slots. 3. Attendant selects a slot. 4. System verifies that the slot is still available. 5. System assigns the slot and marks it Occupied.
Alternate Flow	Slot Taken Before Confirmation

	At Step 4, if the selected slot becomes unavailable: 4a. System prompts the attendant to select another slot. No Compatible Slot At Step 2, if no slot matches the vehicle type: 2a. System displays an error. 2b. Assignment is halted (DR-07).

UC-07 — Track Vehicle

Goal	Locate a parked vehicle and display its current parking slot and status.
Actor	Parking Attendant / Administrator
Main flow	1. Authorized user enters the vehicle number. 2. System searches active parking sessions. 3. System retrieves the matching session. 4. System identifies the assigned parking slot. 5. System displays the vehicle's current location and status.
Alternate Flow	No Active Session Found At Step 3, if no active session matches: 3a. System displays a 'vehicle not currently parked' message. Invalid Vehicle Number At Step 1, if the entered number is blank or malformed: 1a. System displays a validation error and requests a valid number. Unauthorized Access At Step 1, if a non-authorized role attempts access: 1a. System denies access (IR-SEC-02).

UC-08 — Record Vehicle Exit

Goal	Record a vehicle's exit, calculate applicable charges, and close the parking session.
Actor	Parking Attendant
Main flow	1. Attendant enters the vehicle number. 2. System retrieves the active parking session. 3. System records the exit time. 4. System calculates parking duration and fee. 5. Payment is processed. 6. System completes the parking session and releases the slot, marking it Available.
Alternate Flow	No Active Session At Step 2, if no active session exists: 2a. System rejects the exit request. 2b. System displays an error (DR-04).

	Invalid Exit Timestamp At Step 3, if the exit time precedes the entry time: 3a. System displays a validation error. 3b. Session is not completed (DR-09). Payment Fails At Step 5, if payment fails or is declined: 5a. System marks the session as pending payment. 5b. Slot is not released until payment succeeds.

UC-09 — Calculate Parking Fee

Goal	Compute the parking fee for a session based on duration and the configured parking rate.
Actor	System
Main flow	1. System retrieves the entry time. 2. System retrieves the exit time. 3. System calculates the parking duration. 4. System retrieves the configured parking rate. 5. System calculates and displays the fee.
Alternate Flow	Rate Not Configured At Step 4, if no rate is configured for the slot/vehicle type: 4a. System applies the default rate. 4b. System flags the session for admin review. Invalid Duration At Step 3, if the calculated duration is negative or zero due to a data error: 3a. System raises an error instead of generating a fee.

UC-10 — Manage Payment

Goal	Process and record a payment made for a completed parking session.
Actor	Driver / Attendant (Payment Gateway supporting)
Main flow	1. System displays the parking fee. 2. User selects a payment method. 3. Payment request is submitted to the Payment Gateway. 4. Payment gateway processes the transaction. 5. System records the payment result and associates it with the session. 6. System confirms payment.
Alternate Flow	Payment Declined At Step 4, if the payment gateway declines the transaction: 4a. System records a failed status.

	4b. System informs the user. Payment Gateway Unreachable At Step 3, if the gateway does not respond: 3a. System notifies the user. 3b. System allows the payment to be retried later. User Cancels Payment At Step 2, if the user cancels before submission: 2a. System keeps the session marked as unpaid.

UC-11 — View Parking History

Goal	Display a user's historical parking-session information.
Actor	Driver / Administrator
Main flow	1. User opens parking history. 2. System retrieves completed parking sessions. 3. System displays vehicle number, slot, entry/exit times, duration, and fee.
Alternate Flow	No History Available At Step 2, if no completed sessions exist: 2a. System displays an empty-history message. Data Retrieval Error At Step 2, if session data cannot be retrieved: 2a. System displays an error. 2b. System allows the user to retry.

UC-12 — Administration & Reports

Goal	Allow an administrator to manage system data and generate parking-related reports.
Actor	Administrator
Main flow	1. Administrator logs in. 2. Administrator opens administration. 3. Administrator manages users, vehicles, parking slots, and rates. 4. Administrator monitors parking occupancy. 5. Administrator selects the required report. 6. System generates the report.
Alternate Flow	Invalid Configuration Value At Step 3, if an invalid value is entered (e.g., a negative rate): 3a. System rejects the change. 3b. System displays an error. Insufficient Data for Report

	At Step 6, if there is no data for the selected period: 6a. System notifies the administrator instead of generating an incomplete report. Unauthorized Access At Step 2, if a non-admin user attempts access: 2a. System denies access (IR-SEC-08).